

JOHN LI

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EDUCATION

Carnegie Mellon University

Master's of Science in Mechanical Engineering Advanced Study

Concentration: Design and Manufacturing of Mechanical Systems

Pittsburgh, PA

Anticipated Graduation December 2027

Stony Brook University

Bachelor's Degree of Engineering in Mechanical Engineering

Honors: Dean's List (Top 20% of class) | University Scholar Program (Top 8% of the class) | GPA: 3.8

Stony Brook, NY

2022- 2026

ENGINEERING PROJECT & LEADERSHIP

SBU Student Launch Vehicle & Propulsion Subteam Lead | Technical Leadership & Team Management

June 2025 - Present

- Managed full lifecycle development of two custom high-power competition airframes, engineering a Mars rover carrier and a precision altitude vehicle to achieve target apogees (1459 ft and 758 ft) within tight competition margins
- Conducted flight profile simulations using RockSim and MATLAB to optimize vehicle stability, verifying that aerodynamic center and center of gravity margins met structural constraints across the entire flight envelope
- Designed and fabricated robust single-deployment parachute recovery systems, utilizing electronic altimeters and black powder ejection charges to achieve controlled descent rates (12 ft/s and 19 ft/s) and ensure 100% vehicle reusability.
- Served as Technical Lead to direct a 5-member design sub-team and coordinated full system integration across 15+ payload and recovery members, managing manufacturing schedules, design reviews, and safety compliance from PDR through launch day

PROFESSIONAL EXPERIENCES

Stony Brook University Materials and Mechanics Lab

Stony Brook, NY

NASA Space Grant Intern & Undergraduate Research Assistant

May 2025 - Present

- Conducted ASTM D6484 open-hole compression (OHC) testing on woven FRP laminates (hole sizes 3–6 mm) using a Shimadzu UTM and high-speed imaging to characterize load response and fracture process zone (FPZ) damage.
- Analyzed sub-scale failure modes across varying geometries, finding first-damage gross stress converging at 42–46 MPa — isolating a load-introduction-limited failure mode and informing future test protocol improvements.
- Continuing to investigate contributing variables: specimen length, grip clamping conditions, and hole machining quality

Stony Brook University Division of Information Technology

Stony Brook, NY

Senior Lead IT Technician

February 2023 - Present

- Led and mentored a 25+ member technical team, delegating tasks effectively, and promoting a collaborative culture.
- Standardized workflows through SOPs and training programs, enhancing efficiency, team performance, and service quality.
- Directed cross-functional initiatives, aligning technical solutions with organizational goals and enhancing user satisfaction.

Brookhaven National Laboratory (BNL)

Upton, NY

Mechanical Engineering Intern

June 2024 - August 2024

- Utilized Creo and Ansys to model and analyze new structural testing components, ensuring all facility designs met strict engineering margins for the upcoming Electron-Ion Collider (EIC) accelerator program.
- Designed and engineered an innovative end cap and fixture for a critical 591MHz fundamental power coupler (FPC) superconducting radio frequency (SRF) component with guidance of the project manager.
- Prototyped an thermal test rig using additive manufacturing and embedded electronics, expanding facility capabilities to simulate heat transfer across multiple future bellows experiments
- Analyzed experimental thermal data to quantify contact resistance in RF-shielded bellows, directly validating component performance boundaries under simulated operational environments

TECHNICAL SKILLS

Softwares: Ansys, LabVIEW, MATLAB, SolidWorks, Creo, Inventor, Fusion 360, Arduino IDE

Technical: Project Management, Design & Prototyping, Manufacturing Processes (3D Printing & Manual & CNC Machining)

Finite Element Analysis (FEA), Circuit Building & Analysis, Microsoft Office, Google Suite